Practice Exam I

Math 1320 001 Summer 2014
C.Sampson
Julne 20th 2014

Name and Unid: _____

by writing my name I swear by the honor code

Read all of the following information before starting the exam:
Note this is the practice exam ignore point values on problems there are extras but the exam will look similar

- Show all work, clearly and in order, if you want to get full credit. I reserve the right to take off points if I cannot see how you arrived at your answer (even if your final answer is correct).
- Justify your answers algebraically whenever possible to ensure full credit. Sketch all relevant graphs and explain all relevant mathematics.
- Circle or otherwise indicate your final answers.
- You MAY NOT use a calculator or notes. You may use the formulas provided.
- This test has 11 problems and one bonus problem and is worth 100 points, plus some extra credit at the end. It is your responsibility to make sure that you have all of the pages!
- Good luck!

Good Luck!
Here are some Formulas and rules for you.

1. (*5 points*)
Find the volume of the solid obtained by rotating the region bound by the curves $x = 1 + y^2$, $y = x - 3$ about the y-axis

2. (*10 points*)
Find the volume of the solid obtained by rotating the region bound by the curves $x = 2\pi$, $y = sin(x)$, $y = 0$ when rotated around the y-axis.

3. (*10 points*) The force on a particle moving along the x-axis from $x = 0$ to $x = 10$ is given by $F(x) = ln(x)/x$ find the work done on the particle.

4. (*5 points*) A 1600-lb elevator is suspended by a 200-ft cable that weighs 10 lb/ft. How much work is required to raise the elevator from the basement to the third floor, a distance of 30 ft?

5. (*10 points*) You may not have known that I garden in the summer at home. One of my water pails has a small crack in the bottom and leaks as I fill it up from the hose. If the rate of water going into the pail is 0.2 L/ sec and the leak rate is constant at .002 L/s Write a differential equation which describes the rate of change of the volume of water in the pail as I fill it up.

6. (*10 points*) Verify that $y = cos(kx)$ is a solution to the differential equation $y'' + k^2y = 0$ Not don't try to solve just verify.

7. (*10 points*) Verify that $y = -tcos(t) - t$ is a solution to the initial value problem $t\frac{dy}{dt} = y + t^2sin(t)$ with $y(\pi) = 0$ Not don't try to solve just verify.

8. (*5 points*) Sketch A direction field for the equation y'=x/y and use it to sketch solution curves.

9. (*15 points*)

a. (*5 pts*) Solve the initial value problem for the differential equation $\frac{dy}{dx} = x/y$ with y(0)=1

b. (*10 pts*) Find the general solution to the differential equation $e^x h' = sin(x)$

10. (*10 points*) Write the quadratic equation $f(x) = x^2 - 6x + 8$ in standard form.

11. (*20 points*) A population of bacteria grows exponentially and starts with 4 cells. After 20 minuets there are 8 cells.

a. (*5 pts*) Write down the equation that gives the population as a function of time

b. (*5 pts*) When will the population reach 100?

12. (*10 points*)
Suppose that the rate a student learns new mathematics is proportional to the amount of mathematics they know already. Write a differential equation describing the rate that a student learns math. If you started in grade 1 with a math knowledge of 1 and after grade 2 you had a math knowledge of 4 find a function using the differential equation which gives the amount of math you know as a function of time. Consider grade 1 to 2 a time length of one year.

13. (*15 points*) Find the limits of the following sequences if they converge.

a. (*5 pts*) $a_n = \frac{3+5n}{n^4-n^3-n^2}$

b. (*5 pts*) $a_n = \frac{e^n+e^-2n}{e^n-1}$

c. (*5 pts*) $a_n = n^n$

14. (*10 points*) Determine wether the given series is convergent or divergent. If it is convergent find it's sum

a. (*5 pts*) $\sum_{n=1}^{\infty} \frac{1}{\pi}^n + \frac{1}{n} - \frac{1}{n+1}$

b. (*5 pts*) $\sum_{n=1}^{\infty} \frac{n^2+3n^5-n^4}{n^3-n^2+n^5}$

c. (*5 pts*) $\sum_{n=1}^{\infty} \frac{ln(n)}{n}$

15. (*10 points*) Determine wether or not the given series is absolutely convergent , conditionally convergent , or divergent. $\sum_{n=1}^{\infty}(-1)^n \frac{n}{n+9}$

16. (*10 points*) Determine how many terms you need to ensure that the partial sum of the series $\sum_{n=1}^{\infty} \frac{1}{n^4}$ is correct to with in 0.0001

17. (*10 points*) Find the radius of convergence and interval of convergence for the following series.

a. (*5 pts*) $\sum_{n=1}^{\infty} \frac{(x-4)^n}{n^2}$

b. (*5 pts*) $\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n!}$

c. (*5 pts*) $\sum_{n=1}^{\infty} \frac{(-2)^n (x-1)^n}{n}$

18. (*10 points*) Find a power series for the following functions using the geometric series. $\sum_{n=1}^{\infty} x^n = \frac{1}{1-x}$ when $|x| < 1$

a. (*5 pts*) $f(x) = \frac{1}{1+x^2}$

b. (*5 pts*) $f(x) = \frac{x}{(1-x^2)^2}$

c. (*5 pts*) $f(x) = \arctan(2x)$

19. (*10 points*)
Find taylor series out to 4 terms for the following functions at the given centers

a. (*5 pts*) $f(x) = \sqrt{x}$ centered at $x = 1$

b. (*5 pts*) $f(x) = \cos(x)$ centered at $x = \pi$

20. (*10 points*)
Calculate the indefinite integral $\int \frac{\sin(x)}{x} dx$ using power series.

21. (*5 points*) (Bonus all or nothing) Prove using taylor series the identity $e^{ix} = \cos(x) + i\sin(x)$. Then show that $e^{i\pi} = -1$

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